

FOUAD MAHMOUD

AI INFRASTRUCTURE & AGENTIC SYSTEMS ENGINEER

Tanta, Gharbia, Egypt | fouadmahmoud281@gmail.com | www.fouadmahmoud.tech | +20 106 624 8326 | github.com/fouadmahmoud281

SUMMARY

AI engineer working at both ends of the stack: operating multi-tenant GPU infrastructure for LLM inference, and building the agentic systems that run on top of it. Experienced in Kubernetes-based GPU cluster management, vLLM serving, and LangGraph multi-agent architectures taken from prototype to production, with a background in Mechatronics & Robotics.

PROFESSIONAL EXPERIENCE

AI Infrastructure Engineer | Princess Nourah bint Abdulrahman University — Datacenter

Jan 2026 – Present · Riyadh, Saudi Arabia

- Operate multi-tenant GPU clusters with namespace-per-tenant isolation, resource quotas, and RBAC.
- Provision datacenter infrastructure as code with Terraform and Ansible, from bare metal through CIS-hardened Kubernetes.
- Serve LLM inference on vLLM with tensor parallelism behind an OpenAI-compatible API.
- Instrument the estate with Prometheus, DCGM, and Grafana — utilisation heatmaps, NVLink health, and thermal alerting.
- Enforce in-country data residency, secrets management, and admission policy across every workload.

AI Consultant | Sphinx Solutions

Aug 2025 – Present · United Kingdom (Remote)

- Architecture review and feasibility assessment for LLM and agentic initiatives.
- Retrieval strategy and evaluation design, establishing what to measure before committing to a model.
- Cost and model-routing guidance to keep inference spend proportional to value delivered, plus technical mentoring for teams adopting agentic patterns.

AI & Agentic Systems Engineer | Obelion.AI

May 2024 – Present · Remote

- Lead engineer across the Syntera suite: GPU control plane, AI marketplace, and conversational software-engineering studio.
- Architect LangGraph multi-agent systems with typed state and explicit control flow.
- Own retrieval end to end — hybrid dense/sparse search over Qdrant, reranking, and groundedness evaluation.
- Ship FastAPI services and Next.js surfaces, instrumented with tracing, cost accounting, and regression evals.

Scrum Master | OPEX — Cloud Gate

Jan 2024 – May 2024 · Hybrid

- Facilitated agile development of Cloud Gate Marketplace, a multi-cloud automation platform.
- Coordinated cross-functional teams and streamlined sprint planning, backlog management, and communication to improve delivery speed.

AI & Data Science Instructor | Techy School

Feb 2022 – Dec 2023 · Hybrid

- Taught applied AI, LLMs, and data science fundamentals through hands-on projects.
- Developed structured curricula on Python and ML, mentoring 50+ students on real-world AI applications.

FEATURED PROJECTS

- **Syntera Engine** — Multi-datacenter GPU management platform — multi-tenant isolation, vLLM inference orchestration, and full cluster lifecycle on RKE2/Rancher.
- **Syntera Web (Studio)** — Conversational AI software engineering platform automating the SDLC via LangGraph agents, from requirements through to pull request.
- **Syntera Marketplace** — AI marketplace and control plane for LLM endpoints, vector databases, and deployable agentic workflows.
- **Bayan** — Bilingual (AR/EN) AI librarian with LangGraph retrieval over Qdrant hybrid search, character chat, and synthesised audio summaries.
- **Inpakt** — Bilingual social impact tracking platform with agentic project setup, OCR document analysis, and AI-generated narrative reporting.

EDUCATION

Bachelor of Engineering in Mechatronics & Robotics | Tanta University

Sep 2020 – May 2026 · Tanta, Egypt

Relevant coursework in Computer Vision, Programming, and Autonomous Agents.

TECHNICAL SKILLS

AI & Agents: LangChain, LangGraph, LangSmith, Langfuse, OpenAI API, RAG, multi-agent architecture, evaluation & groundedness scoring

Infrastructure: Kubernetes (RKE2), Rancher, Cilium/eBPF, NVIDIA GPU Operator, DCGM, MIG, vLLM, Terraform, Ansible, Prometheus, Grafana, Vault, Harbor

Backend & Web: Python, FastAPI, Flask, TypeScript, Next.js, React, Docker, CI/CD

Data & ML: Qdrant, PostgreSQL, MongoDB, Redis, MinIO, PyTorch, Scikit-learn, Streamlit, data pipelines

ADDITIONAL INFORMATION

Languages: English, Arabic, German.

Interests / Research: Agentic AI systems, AI infrastructure at scale, entrepreneurial AI solutions, product innovation.